Figure 12: Layer-wise PCA visualization (Component 1 vs. Component 2) for Gemma-2-9B-Instruct on two complex datasets: **`common_claim_true_false`** (top row) and **`counterfact_true_false`** (bottom row). Columns represent different instructional conditions: Neutral, Truthful, and Deceptive prompts. Visualizations are performed on **Layer 16**, the key layer identified for Gemma-2-9B-Instruct based on the probing accuracy peaks in Figure 3.

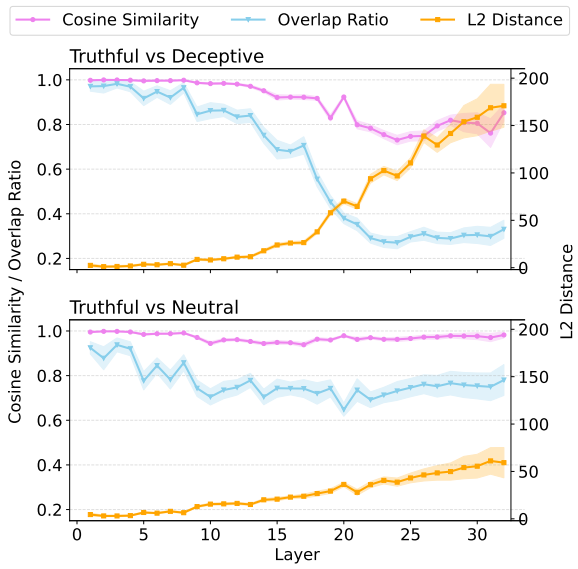


Figure 13: Layer-wise feature shift analysis for Gemma-2-9B-Instruct on `common_claim_true_false`.

D SAE-based Layer-wise Feature Shift Analysis

We analyze how sparse feature activations shift across layers under different instruction types using three metrics in SAE latent space: Cosine Similarity, Overlap Ratio, and L2 Distance. Figures 13 and 14 show results for the `common_claim_true_false` and `counterfact_true_false` datasets using Gemma-2-9B-Instruct.

In both datasets, deceptive prompts induce strong mid-to-late layer shifts, especially between Layers 16 and 32. This is evidenced by the sharp rise in L2 distance and the corresponding drop in cosine similarity and overlap ratio when comparing truthful and deceptive inputs. The effect is most pronounced in `counterfact_true_false`, where overlap sharply declines post-Layer 16, indicating a reconfiguration of sparse feature sets. In contrast, shifts between truthful and neutral prompts remain small and gradual across all layers, suggesting that the major representational changes are deception-specific.

These results highlight a distinctive geometric transformation in the model's latent representations under deceptive instructions and further motivate mid-layer analysis when identifying potential deception-sensitive features. This pattern closely aligns with earlier findings from linear probing, where intermediate layers—especially around Layer 16—also showed peak decodability of the model's intended "True"/"False" output across instruction types.

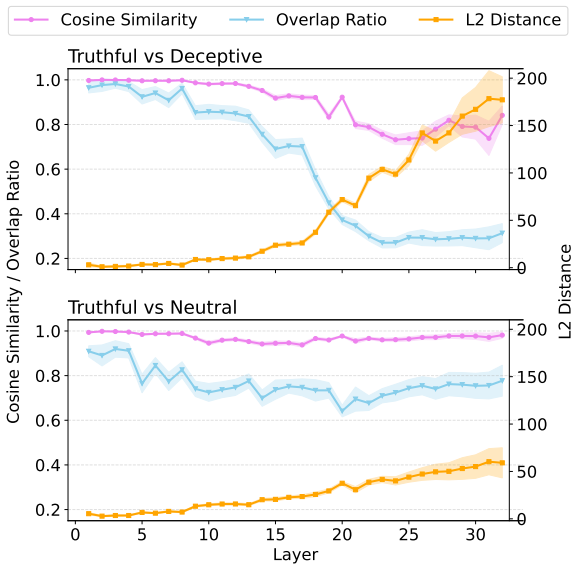


Figure 14: Layer-wise feature shift analysis for Gemma-2-9B-Instruct on `counterfact_true_false`.

E SAE-based Neuron-wise Feature Shift Analysis

This appendix section provides additional visualizations to support the neuron-wise feature shift analysis detailed in main text. The core objective of this analysis is to move beyond global representational shifts and pinpoint specific SAE features that are most sensitive to the change from truthful to deceptive instructions. By examining individual SAE feature activations, we can gain a better understanding of how deception is encoded at the feature level.

The methodology involves identifying, for each layer, the sparse SAE features exhibiting the largest change in average activation when comparing the Deceptive condition to the Truthful condition. The figures presented in this appendix, such as scatter plots showing the activation of the top distinguishing features (similar to Figure 7 for `common_claim_true_false` but potentially for other datasets like `counterfact_true_false` as shown in Figure 15 and violin plots in Figure 16 illustrating the distribution of these feature activations under truthful versus deceptive prompts, offer further evidence.

These supplementary visualizations help to reinforce the finding that a small subset of sparse features often displays a near-binary activation pattern—being highly active for one instruction type (e.g., truthful) and suppressed for the other (e.g., deceptive), or vice-versa. This detailed view corroborates the idea that these specific features act as "deception-associated features," playing a critical role in modulating the model's internal representation in response to deceptive instructions, often aligning with the mid- and late-layer SAE shift peaks identified globally in Figure 6 and Figure 7. The plots here may cover additional layers or datasets, providing a more comprehensive picture of this phenomenon.

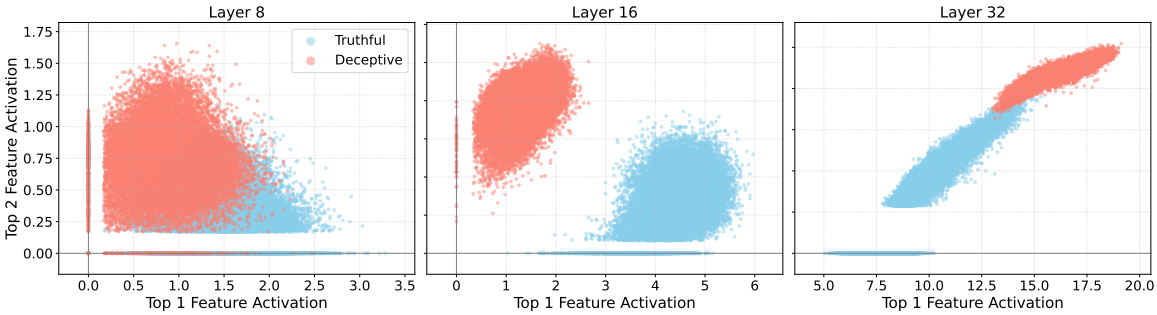
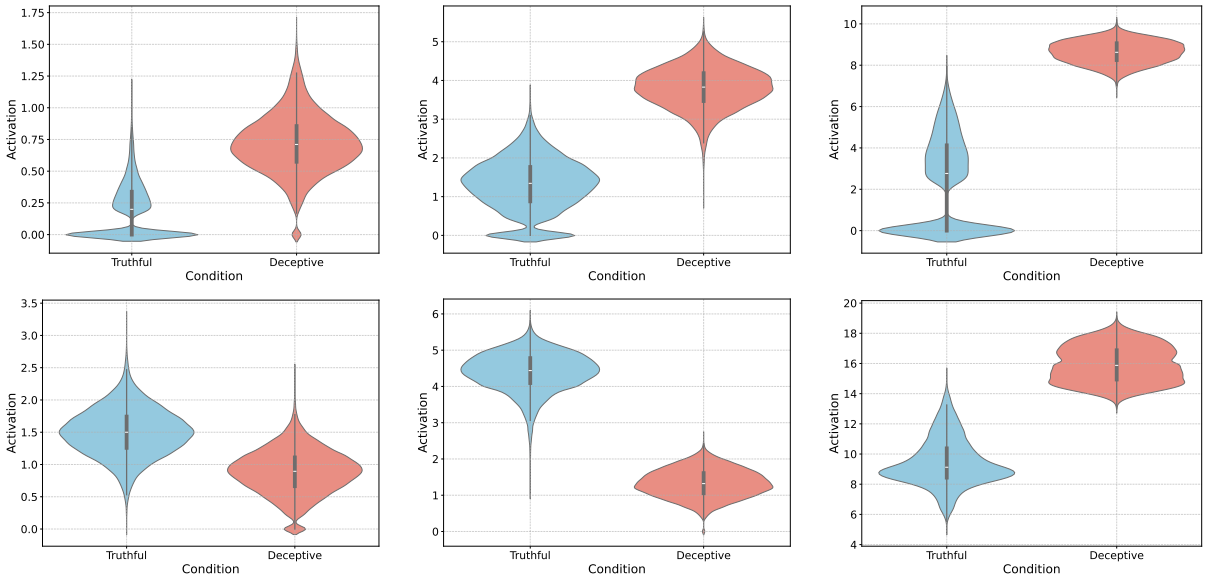


Figure 15: Neuron-by-neuron feature shift analysis for LLaMA-3.1-8B-Instruct on `counterfact_true_false`



(a) Layer 8 (b) Layer 16 (c) Layer 32

Figure 16: Violin graph of LLaMA-3.1-8B-Instruct activations on the `counterfact_true_false`. The top row displays the activation distributions for the SAE feature most responsive to deceptive instructions (Top 1 feature), while the bottom row shows the distributions for the second most responsive feature (Top 2 feature), across layers 8 (a), 16 (b), and 32 (c).